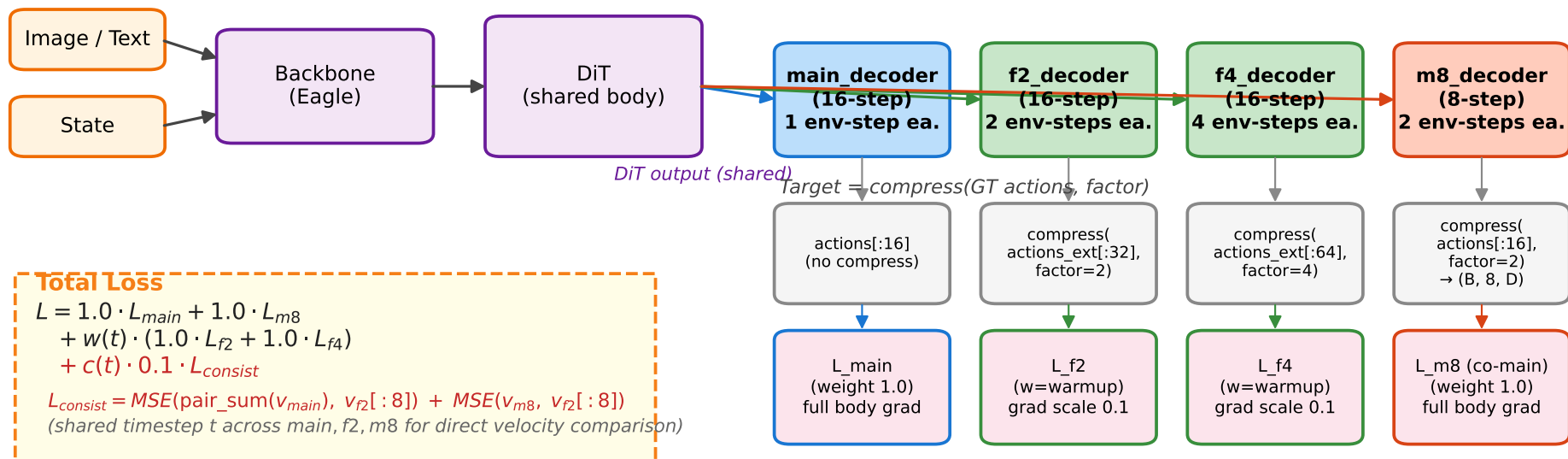
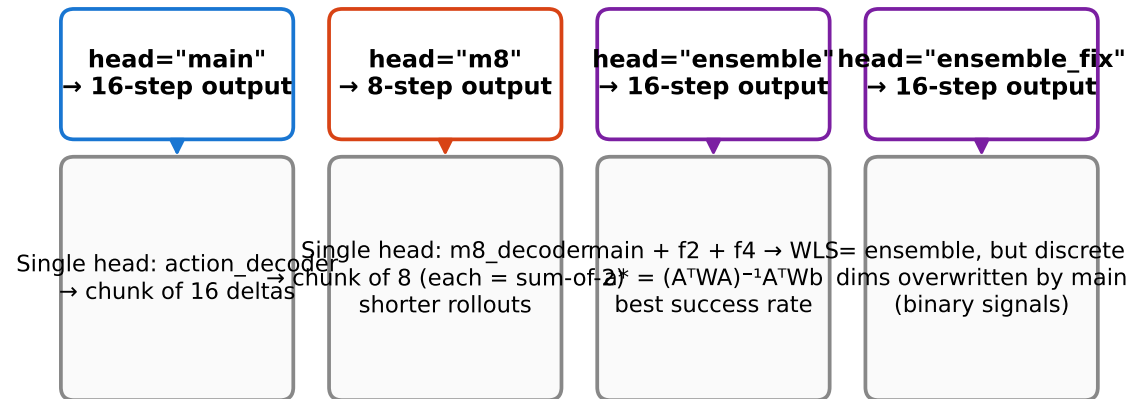


(A) Training: shared backbone/DiT + 4 decoder heads with horizon-specific targets



(B) Inference modes (each runs flow-matching denoising 4 steps)



(C) WLS ensemble (toy: H=4, factor=[2])

Predictions (toy: 4 unknowns $a[0..3]$):

main $\mathbf{m} = [m_0, m_1, m_2, m_3]$

f2 $\mathbf{f} = [f_0, f_1]$ (each = sum of 2)

Stacked observations $\mathbf{b}$:

$\mathbf{b} = [m_0, m_1, m_2, m_3, f_0, f_1]^T$

Constraints $\mathbf{A}$, weights $\mathbf{W}$:

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix} \quad \mathbf{W} = \text{diag}(1, 1, 1, 1, \frac{1}{2}, \frac{1}{2})$$

Solve: $\mathbf{a}^* = (\mathbf{A}^T \mathbf{W} \mathbf{A})^{-1} \mathbf{A}^T \mathbf{W} \mathbf{b}$

$$= \arg \min_{\mathbf{a}} \sum_i w_i ((\mathbf{A} \mathbf{a})_i - b_i)^2$$